

Vamsikrishna Chemudupati

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SUMMARY

AI/ML Engineer with 6+ years of experience developing and deploying production-grade AI solutions specializing in NLP, ASR, and LLM-based applications across cloud and edge environments.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL, Shell

ML/LLM Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, ONNX Runtime, vLLM, llama.cpp

Cloud & Infrastructure: Azure, AWS, Microsoft Agent Framework, Strands Agent SDK

Tools: Git, Infrastructure-as-Code (IaC), Apache Airflow, FastAPI, Docker, CI/CD, ElasticSearch

WORK EXPERIENCE

Lead AI Engineer

Bank of Montreal, Toronto, Canada

September 2025 - Present

- Developed a conversational AI agent leveraging RAG and Text-to-SQL to audit vendor contracts against ServiceNow license data, identifying compliance gaps and mitigating up to \$7M in annual licensing penalties.
- Implemented a ReAct-based multi-agent system integrated into BMO Investor Platform (500k users), using tool calling for real-time portfolio analysis, account balance lookup, knowledge retrieval, and automated meeting scheduling.
- Led the production deployment of agentic workflows on Azure platform, defining the solution architecture, infrastructure, observability and security requirements to support enterprise-wide adoption.
- Designed and deployed a code-execution runtime for the AI agents, enabling secure execution of LLM-generated code across deterministic and non-deterministic agentic workflows.
- Provisioned Azure & AWS environments utilizing IaC & Terraform involving granular role-based access controls, data governance, and compute isolation to support secure multi-agent workloads.

Sr. Machine learning Engineer

Avid Technology, Toronto, Canada

August 2023 - July 2025

- Developed Speaker Diarization solution using a combination of Whisper, speaker embedding model (ResNet34) and multi-stage clustering method to handle varying audio lengths. Achieved a Diarization error rate of 7.98% when evaluated on conversational datasets.
- Worked with fine-tuning Whisper model (Seq-Seq) for low resource languages and accents using LoRA method to improve WER metric.
- Deployed 8-bit quantized LLMs (Gemma 3) on edge and server-based platforms using llama.cpp and vLLM inference engines for downstream tasks such as transcript translation and summarization.
- Led the deployment of diarization solution on windows and mac edge devices using ONNX Runtime library and C++. Integrated an Elasticsearch Vector DB to facilitate the metadata storage of transcripts and speaker identities.

NLP Researcher

Huawei Technologies (Noah's Ark Lab), Montreal, Canada

January 2023 - August 2023

- Working on developing Web-assisted QA systems using RAG methods with MongoDB as the document database. Developed data processing pipelines involving web scrapping and data cleaning from publicly available chatbots, resulting in 500k QA samples. Performed instruction-based training on GPT based LLM's (LLaMa 7B, 13B) using the multi-domain data collected.
- Created an automated solution for evaluating the predicted answer using LLM's fine-tuned on Entailment and paraphrase detection modules.
- Evaluated the robustness and cross-task performance of ASR system Whisper by implementing a data pipeline that corrupts input speech using environment noise and room reverberations. Achieved SOTA results in both clean and noisy environments across multiple ASR tasks given in SUPERB benchmark, indicating the potential of deployment in real-time products.

Research Scientist Intern

Nuance Communications (Microsoft Company), Montreal, Canada

May 2022 - December 2022

- Developed a single End-to-End ASR model with configurable accuracy/latency trade-off at runtime using Universally Slimmable techniques and in-place distillation techniques.
- Implemented channel-wise and depth-wise slimming techniques on a Conformer-based RNN-T model using TensorFlow, achieving a configurable accuracy-latency tradeoff with only 1-2% relative WER degradation compared to dedicated baseline models of equivalent size.

EARLIER EXPERIENCE

Machine Learning Engineer, Hexaware Technologies

Dec 2020 – Aug 2021

Research Engineer, Hyundai Mobis

Jul 2018 – Nov 2020

EDUCATION

University of Montreal | Mila Lab

Montreal, Quebec, Canada

Master of Science: Computer Science (Machine Learning)

May 2023

Vellore Institute of Technology

Vellore, Tamil Nadu, India

Bachelor of Technology: Electronics and Communication

May 2018

PUBLICATIONS

1. **Vamsikrishna C.**, Tahaei M. et al. “On the Transferability of Whisper-based Representations for ‘In-the-Wild’ Cross Task Downstream Speech Applications.” in *arXiv*, 2023. arxiv.org/pdf/2305.14546
2. Kaushik M., Ankit R., **Vamsikrishna C.**, and Xing Han. “TASTY: A Transformer based approach to space and time complexity.” in *International Conference on Learning Representations (ICLR) Deep Learning for Code Workshop*, 2023. arxiv.org/abs/2305.05379